RHC-UCRL

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1 Problem Statement

We assume a stochastic environment where the state space denoted as $\mathcal{S} \subseteq \mathbb{R}^p$ and the action space is denoted as $\mathcal{A} \subseteq \mathbb{R}^q$. The adversary actions are denoted as $\bar{\mathcal{A}} \subseteq \mathbb{R}^q$ At any state $s_h \in \mathcal{S}$ the transition dynamics is defined by function $f : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}}$. Both the action sets considered to be compact and the dynamics are given by

$$s_{h+1} = f(s_h, a_h, \bar{a}_h) + \omega_h \tag{1}$$

The reward and utility functions are given by $r : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to \mathbb{R}$ and $u : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to [0, 1]$ respectively.

Algorithm 1 RHC-UCRL

Input: intital state s_0, reward function $r : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to \mathbb{R}$, utility function $u : \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}} \to [0, 1]$, horizon length H, Iteration length τ,dual parameter λ initial policies $\pi_0, \bar{\pi}_0$ (Check if required)
for $t = 1, 2, \ldots, \tau$ **do**
    Reset the system to $s_{0,t+1} = s_0$
    $\pi_t \in \Pi$(Optimistic policy pick for agent) (Complete this)
    $\bar{\pi}_t \in \bar{\Pi}$ (Pessimistic policy pick for adversary) (Complete this)
    for $h = 1, \ldots, H$ **do**
        $s_{(h,t)} = f(s_{(h-1,t)}, \pi(a_{(h-1,t)}|s_{(h-1,t)}), \bar{\pi}_t(\bar{a}_{(h-1,t)}|s_{(h-1,t)})) + \omega_{h,t}$
        Collect $\big(s_{(h-1,t)}, a_{(h-1,t)}, \bar{a}_{(h-1,t)}, s_{(h,t)}, r(s_{(h-1,t)}, a_{(h-1,t)}, \bar{a}_{(h-1,t)}), u(s_{(h-1,t)}, a_{(h-1,t)}, \bar{a}_{(h-1,t)})\big)$
    end for
    Solve $\pi_t \in \arg\max_{\pi \in \Pi} \min_{\bar{\pi} \in \bar{\Pi}} \left(J^{(o)}(\pi, \bar{\pi}) - \lambda \left[b_n - J^{(o)}_{u_n}(\pi, \bar{\pi}) \right]_+ \right)$
    $\bar{\pi}_t \in \arg\min_{\bar{\pi}} \left(J^{(p)}(\pi_t, \bar{\pi}) - \lambda \left[b_n - J^{(p)}_{u_n}(\pi_t, \bar{\pi}) \right]_+ \right)$
end for

Assumption 1. *The function f that determines the dynamics in equation* (1) *is L_f-Lipschitz continuous and the noise ω_h $\forall$ $h \in \{0, 1, \ldots, H-1\}$ are i.i.d. σ-Sub-Gaussian.*

Let us consider $J(f, \pi, \bar{\pi})$ and $J_u(f, \pi, \bar{\pi})$ as the expected long-run reward and the expected long-run utility value. These estimates evaluate the performance of a pair of polices $(\pi, \bar{\pi})$. The evaluations are performed on episodic data collected by playing the policy pair under evaluation. Formally, they can be defined as equation (2) and (3)

$$J(\tilde{f}, \pi, \bar{\pi}) := \mathbb{E}_{\nu_{\tilde{f},\pi,\bar{\pi}}} \left[\sum_{h=0}^{H} r(\mathbf{s}_h, \mathbf{a}_h, \bar{\mathbf{a}}_h) | \mathbf{s}_0 \right], \tag{2}$$

$$J_u(\tilde{f}, \pi, \bar{\pi}) := \mathbb{E}_{\nu_{\tilde{f},\pi,\bar{\pi}}} \left[\frac{1}{H} \sum_{h=0}^{H-1} u(\mathbf{s}_h, \mathbf{a}_h, \bar{\mathbf{a}}_h) | \mathbf{s}_0 \right], \tag{3}$$

such that the state transition is given by $\mathbf{s_{h+1}} = \tilde{f}(\mathbf{s}_h, \mathbf{a}_h, \bar{\mathbf{a}}_h) + \omega_h$

The problem that we aim to solve is as mentioned in equation (4)

$$\begin{aligned} &\max_{\pi \in \Pi} \min_{\bar{\pi} \in \bar{\Pi}} J(f, \pi, \bar{\pi}), \\ &\textbf{s.t.} \ \min_{\bar{\pi} \leq \bar{\Pi}} J_u(s, \pi, \bar{\pi}) \geq b. \end{aligned} \tag{4}$$

Thus, we aim to find π^* defined as

$$\pi^* \in \arg \min_{\lambda \geq 0} \max_{\pi \in \Pi} \min_{\bar{\pi} \in \bar{\Pi}} J(f, \pi, \bar{\pi}) + \lambda(\min_{\bar{\pi} \in \bar{\Pi}} J_u(f, \pi, \bar{\pi}) - b). \tag{5}$$

To solve Algorithm 1, we need $J_t^{(o)}(\pi, \bar{\pi})$ and $J_t^{(p)}(\pi, \bar{\pi})$ defined in equations (6) and (7) respectively.

$$\begin{aligned} &J_t^{(o)}(\pi, \bar{\pi}) := \max_{\eta^{(o)}} J(f^{(o)}, \pi, \bar{\pi}), \\ &\text{s.t. } f^{(o)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) = \mu_{t-1}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) + \beta_t \eta^{(o)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) \Sigma_{t-1}^{1/2}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}). \end{aligned} \tag{6}$$

$$\begin{aligned} &J_t^{(p)}(\pi, \bar{\pi}) := \min_{\eta^{(o)}} J(f^{(p)}, \pi, \bar{\pi}), \\ &\text{s.t. } f^{(p)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) = \mu_{t-1}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) + \beta_t \eta^{(p)}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}) \Sigma_{t-1}^{1/2}(\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}}). \end{aligned} \tag{7}$$

Performance metric: The aim is to find a policy π_T such that $\min_{\bar{\pi}} J(f, \pi_T, \bar{\pi}) \geq \min_{\bar{\pi}} J(f, \pi^*, \bar{\pi}) - \epsilon$ and $\min_{\bar{\pi}} J(f, \pi_T, \bar{\pi}) - b < \xi$

Statistical Model: We consider a model based RL approach. That is the goal is to learn the dynamics and find a near-optimal robust policy. We use statistical estimation to probabilistically reason about dynamic models $\tilde{f}$ that are compatible with the observed data $\mathcal{D}_{1:t} = \{\nu_{f,\pi_{t'},\bar{\pi}_{t'}}\}_{t'=1}^{t}$. This can be done by frequentist method or the bayesian perspective.

Assumption 2. *The statistical model is calibrated w.r.t f in* (1)*, so that with $\sigma_t(.) = diag(\Sigma_t(.))$ and a non-decreasing sequence of parameters $\{\beta_t\}_{t \geq 1} \in \mathbb{R}_+$, each depending on $\delta \in (0, 1)$, it holds jointly for all $t \geq 1$ and $\mathbf{s}, \mathbf{a}, \bar{\mathbf{a}} \in \mathcal{S} \times \mathcal{A} \times \bar{\mathcal{A}}$*

such that $|f(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})-\mu_{t-1}(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})| \leq \beta_t\sigma_{t-1}(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})$ *elementwise with probability at least* $1-\delta$

The calibrated model assumption (Assumption 2) is very common in literature [2]. This assumption is crucial for exploration: if the model is not well-calibrated, then leveraging its epistemic uncertainty does not provably guide exploration. For dynamics with bounded norm in a known RKHS, Assumption 2 is satisfied [5, 1]. In the case of neural network models, one-step-ahead predictions can be recalibrated [3].

We construct the set of plausible models at time t as

$$\mathcal{M}_t = \left\{ \tilde{f} \ : \ \left\| \tilde{f}(\cdot) - \mu_{t-1}(\cdot) \right\| \leq \beta_t \, \sigma_{t-1}(\cdot) \right\}.$$

By Assumption 2, we can guarantee that, with high probability, the true dynamics $f \in \mathcal{M}_t$ for all time steps t.

Assumption 3. *For every episode* t*, the functions* Σ_t*, the agent's policy* $\pi_t \in \Pi$*, the adversary's policy* $\bar{\pi}_t \in \bar{\Pi}$*, the reward function* $r(.,.,.)$ *and the utility function* $u(.,.,.)$ *are Lipschitz continuous with respective constants as* $L_\sigma, L_\pi, L_{\bar{\pi}}, L_r$ *and* L_u*.*

Assumption 4. *For all* $\mathbf{s} \in \mathcal{S}$*,* $\mathbf{a} \in \mathcal{A}$ *and* $\bar{\mathbf{a}} \in \bar{\mathcal{A}}$*, we assume that* $|r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}})| \leq R_{max}$

Assumption 4 is a very common practice in RL literature. Mostly the reward is normalized to lie in $[0,1]$, which can be further scaled if required.

Assumption 5. *In general the problem defined in equation* (5) *does not have strong duality guarantees. However, in this work we assume the guarantees for tabular setting and that strong duality holds. There exists a policy* π^* *which uniquely solves equation* (5) *for all state transition function* $f \in \mathcal{F}$ *where* $\mathcal{F} \subseteq \mathcal{M}_F$ *is the set of all such functions.*

2 Proofs

Lemma 1. *(Adapted from Corollary 1 in [4]) Based on assumptions 1 and 3, for every* $\mathbf{s}, \mathbf{s}^{'} \in \mathcal{S}$ *the following ineqlity holds*

$$\|f(\mathbf{s},\pi(\mathbf{s}),\bar{\pi}(\mathbf{s})) - f(\mathbf{s}^{'},\pi(\mathbf{s}^{'}),\bar{\pi}(\mathbf{s}^{'}))\| \leq L_f\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\|\mathbf{s}-\mathbf{s}^{'}\|_2. \tag{8}$$

Proof.

$$\begin{aligned}\|f(\mathbf{s},\pi(\mathbf{s}),\bar{\pi}(\mathbf{s})) - f(\mathbf{s}^{'},\pi(\mathbf{s}^{'}),\bar{\pi}(\mathbf{s}^{'}))\| &\leq L_f\sqrt{\|\mathbf{s}-\mathbf{s}'\|_2^2+\|\pi(\mathbf{s})-\pi(\mathbf{s}')\|_2^2+\|\bar{\pi}(\mathbf{s})-\bar{\pi}(\mathbf{s}')\|_2^2}, \\ &\leq \sqrt{\|\mathbf{s}-\mathbf{s}'\|_2^2+L_\pi^2\|\mathbf{s}-\mathbf{s}'\|_2^2+L_{\bar{\pi}}^2\|\mathbf{s}-\mathbf{s}'\|_2^2} \\ &= L_f\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\|\mathbf{s}-\mathbf{s}^{'}\|_2\end{aligned} \tag{9}$$

The first inequality holds due to the Lipschitz continuity of f. The second inequality is due to Lipschitz continuity of π and $\bar{\pi}$, which we assume in Assumptions 1 and 3 □

Lemma 2. *(Adapted from Lemma 3 in [4]). Assuming the assumptions 1 and 3 are true, the following inequality holds:*

$$|J(f,\pi,\bar{\pi}) - J(\tilde{f},\pi,\bar{\pi})| \leq L_r\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|\right] \tag{10}$$

Proof.

$$\begin{aligned}
|J(f,\pi,\bar{\pi}) - J(\tilde{f},\pi,\bar{\pi})| &= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - \sum_{h=0}^{H-1} r(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} (r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - r(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\sum_{h=0}^{H-1}\mathbb{E}\left[(r(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - r(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&\leq L_r\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|_2\right]
\end{aligned} \tag{11}$$

□

Lemma 3. *Assuming that assumptions 1 and 3 holds, the following inequality is true:*

$$|J_u(f,\pi,\bar{\pi}) - J_u(\tilde{f},\pi,\bar{\pi})| \leq L_u\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|\right] \tag{12}$$

Proof.

$$\begin{aligned}
|J_u(f,\pi,\bar{\pi}) - J_u(\tilde{f},\pi,\bar{\pi})| &= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} u(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - \sum_{h=0}^{H-1} u(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\mathbb{E}\left[\sum_{h=0}^{H-1} (u(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - u(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&= \left|\sum_{h=0}^{H-1}\mathbb{E}\left[(u(\mathbf{s},\mathbf{a},\bar{\mathbf{a}}) - u(\tilde{\mathbf{s}},\tilde{\mathbf{a}},\tilde{\bar{a}})\right]\right| \\
&\leq L_u\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h - \tilde{\mathbf{s}}_h\|_2\right]
\end{aligned} \tag{13}$$

□

Lemma 4. *Assuming that assumptions 1, 2 and 3 are true, then from Lemma 3 following result holds*

$$|\min_{\bar{\pi}\in\bar{\Pi}} J_u(f,\pi,\bar{\pi}) - \min_{\bar{\pi}\in\bar{\Pi}} J_u(\tilde{f},\pi,\bar{\pi})| \leq L_u\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h-\tilde{\mathbf{s}}_h\|_2\right] \quad (14)$$

Proof.

$$\begin{aligned}|\min_{\bar{\pi}\in\bar{\Pi}} J_u(f,\pi,\bar{\pi}) - \min_{\bar{\pi}\in\bar{\Pi}} J_u(\tilde{f},\pi,\bar{\pi})| &\leq \sup_{\bar{\pi}\in\bar{\Pi}} |J_u(f,\pi,\bar{\pi}) - J_u(\tilde{f},\pi,\bar{\pi})| \\ &\leq L_u\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\sum_{h=0}^{H-1}\mathbb{E}\left[\|\mathbf{s}_h-\tilde{\mathbf{s}}_h\|_2\right]\end{aligned} \quad (15)$$

□

Lemma 5. *Under assumptions 1, 2 and 3, for all episodes $t \geq 1$, any $\eta \in [-1,1], h \in \{0,\ldots,H-1\}, \pi \in \Pi$ and $\bar{\pi} \in \bar{\Pi}$ the following inequality holds:*

$$\|\boldsymbol{s}_{h,t}-\tilde{\boldsymbol{s}}_{h,t}\| \leq 2\beta_{t-1}\left((1+L_f+2\beta_{t-1}L_\sigma)\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\right)^{h-1}\sum_{h'=0}^{h-1}\left\|\sigma_{t-1}\big(s_{h',t},\pi_t(s_{h',t}),\bar{\pi}_t(s_{h',t})\big)\right\|_2. \quad (16)$$

Proof. To avoid notational complexity, we use $L_{f,\pi} = L_f\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}$ and $L_{\sigma,\pi} = L_\sigma\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}$.

We prove by induction that

$$\|s_{h,t}-\tilde{s}_{h,t}\|_2 \leq 2.\beta_{t-1}\sum_{h'=0}^{h-1}(L_{f,\pi}+2.\beta_{t-1}L_{\sigma,\pi})^{h-1-h'}.\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h',t})\| \quad (17)$$

For $h=0$, clearly $s_{0,t} = \tilde{s_{0,t}}$. This is because we treat our initial state as fixed. We know, $s_{h+1,t} = f(s_{h,t},\pi_t,\bar{\pi}_t)$. Thus, we can expand the LHS for $h+1$ as:

$$\begin{aligned}\|\mathbf{s}_{h+1,t}-\tilde{\mathbf{s}}_{h+1,t}\| &= \|f(\mathbf{s}_{h,t},\pi_t,\bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t)\| \\ &= \|f(\mathbf{s}_{h,t},\pi_t,\bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t) + f(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t) - f(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t)\| \\ &\leq \|f(\mathbf{s}_{h,t},\pi_t,\bar{\pi}_t) - f(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t)\| + \|f(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t) - \tilde{f}(\tilde{\mathbf{s}}_{h,t},\pi_t,\bar{\pi}_t)\| \\ &\leq L_f\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\|\mathbf{s}-\tilde{\mathbf{s}}\|_2 + 2\beta_{t-1}\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\tilde{\mathbf{s}}_{h,t})\|_2 \\ &= L_{f,\pi}\|\mathbf{s}_{h,t}-\tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1}\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\tilde{\mathbf{s}}_{h,t}) + \sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t}) - \sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 \\ &\leq L_{f,\pi}\|\mathbf{s}_{h,t}-\tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1}(\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\tilde{\mathbf{s}}_{h,t}) - \sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 + \|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t})\|_2) \\ &\leq L_{f,\pi}\|\mathbf{s}_{h,t}-\tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1}L_\sigma\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\|\mathbf{s}_{h,t}-\tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1}\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 \\ &= (L_{f,\pi}+2\beta_{t-1}L_{\sigma,\pi})\|\mathbf{s}_{h,t}-\tilde{\mathbf{s}}_{h,t}\|_2 + 2\beta_{t-1}\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t})\|_2 \\ &\leq 2\beta_{t-1}\sum_{h'=0}^{(h+1)-1}\left((L_f+2\beta_{t-1}L_\sigma)\sqrt{1+L_\pi^2+L_{\bar{\pi}}^2}\right)^{(h+1)-1-h'}\left\|\sigma_{t-1}^{\pi_t,\bar{\pi}_t}(\mathbf{s}_{h,t})\right\|_2\end{aligned} \quad (18)$$

Finally, notice that $(L_{f,\pi}+2\beta_{t-1}L_{\sigma,\pi})^{(h-1-h')} < (1+L_{f,\pi}+2\beta_{t-1}L_{\sigma,\pi})^{h-1-h'} \leq (1+L_{f,\pi}+2\beta_{t-1}L_{\sigma,\pi})^{h-1}$ and the result follows when the above inequality is combined with final inequality from equation (18). □

Lemma 6. *Assuming assumption 4 holds, π_t is the chosen policy at instant t by Algorithm 1 and $c > 2$, the following inequality holds*

$$[b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ \leq cR_{\max}/\lambda \tag{19}$$

Proof.

$$\lambda[b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ \leq \min_{\bar{\pi}} J_r^{(o)}(\pi_t, \bar{\pi}) + \lambda[b - J_u^{(o)}(\pi_t, \bar{\pi})]_+ - \min_{\bar{\pi}} J_r^{(o)}(\pi^*, \bar{\pi}) - \lambda[b - J_u^{(o)}(\pi^*, \bar{\pi})]_+ + cR_{\max}$$
$$\leq cR_{\max} \tag{20}$$

The first inequality is true because π^*, being the optimal policy has returns the highest value. Thus, $[b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ \leq cR_{\max}/\lambda$ □

Lemma 7. *Assuming assumptions 2 and 4 holds,*$\alpha = \frac{cR_{\max}}{\lambda.2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\left(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\right)\|_2\right]}$, *$\pi^*$ be the optimal policy solving equation (4) and π_t and $\bar{\pi}_t$ be the policies selected by Algorithm 1 at instant t, then the following inequality holds*

$$[b - J_u(f, \pi_t, \bar{\pi}_t)]_+ \leq 2L_u H\beta_T^H C^H(1+\alpha) \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\left(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\right)\|_2\right]. \tag{21}$$

Proof.

$$
\begin{aligned}
&[b - J_u(f, \pi_t, \bar{\pi}_t)]_+ = [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}) + \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}) - J_u(f, \pi_t, \bar{\pi}_t)]_+ \\
&\leq [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ + [\min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi}) - J_u(f, \pi_t, \bar{\pi}_t)]_+ \\
&\leq [b - \min_{\bar{\pi}} J_u^{(o)}(\pi_t, \bar{\pi})]_+ + |J_u^{(o)}(\pi_t, \bar{\pi}_t) - J_u(f, \pi_t, \bar{\pi}_t)| \\
&\leq cR_{\max}/\lambda + L_u \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \sum_{h=0}^{H-1} \mathbb{E}\left[\|s_h - \tilde{s}_h\|_2\right] \\
&\leq cR_{\max}/\lambda + L_{u,\pi} \sum_{h=0}^{H-1} \mathbb{E}\left[2\beta_{t-1}\left((1 + L_f + 2\beta_{t-1}L_\sigma)\sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2}\right)^{h-1} \sum_{h'=0}^{h-1} \left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right| \right. \\
&\leq cR_{\max}/\lambda + 2L_u\beta_T^H C^H \sum_{h=0}^{H-1} \mathbb{E}\left[\sum_{h'=0}^{h-1} \left\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\right\|_2\right] \\
&\leq cR_{\max}/\lambda + 2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right] \\
&= 2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right] \left(\frac{cR_{\max}.}{\lambda.2L_u H\beta_T^H C^H \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_h\right.}\right. \\
&= 2L_u H\beta_T^H C^H (1+\alpha) \sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]
\end{aligned}
\tag{22}
$$

This completes the proof. □

Lemma 8. *Let π^* be the optimal feasible policy and the solution for equation 4 and let π_t and $\bar{\pi}_t$ be the polices selected by $RHC-UCRL$ at time t. Then under assumption 2 and $L_{(r,\lambda,u)} = (L_r + \lambda L_u)$, the following inequality (equation (23)) hold with probability at least $1 - \delta_r$.*

$$
\min_{\bar{\pi}} J(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}} J(f, \pi_t, \bar{\pi}) \leq 4L_{(r,\lambda,u)}\beta_T^H C^H H \sum_{h'=0}^{h-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]. \tag{23}
$$

Proof. Let us consider $\min_{\bar{\pi}} J(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}} J(f, \pi_t, \bar{\pi})$ as the instantaneous robust regret of the selected policy π_t and the quantity $\min_{\bar{\pi}} J_u(f, \pi_t, \bar{\pi}) - b$ as the instantaneous robust violation of the same policy. We first determine the

instantaneous regret bound, followed by the instantaneous violation bound.

$$
\begin{aligned}
&\min_{\bar{\pi}} J(f,\pi^*,\bar{\pi}) - \min_{\bar{\pi}} J(f,\pi_t,\bar{\pi}) \le \min_{\bar{\pi}} J_t^{(o)}(\pi^*,\bar{\pi}) - \min_{\bar{\pi}} J(f,\pi_t,\bar{\pi}) \\
&\le \min_{\bar{\pi}} \left(J_t^{(o)}(\pi^*,\bar{\pi}) + \lambda[b - J_u^{(o)}(\pi^*,\bar{\pi})]_+ \right) - \min_{\bar{\pi}} J(f,\pi_t,\bar{\pi}) \\
&\le \min_{\bar{\pi}} \left(J_t^{(o)}(\pi_t,\bar{\pi}) + \lambda[b - J_u^{(o)}(\pi_t,\bar{\pi})]_+ \right) - \min_{\bar{\pi}} J(f,\pi_t,\bar{\pi}) \\
&\le J_t^{(o)}(\pi_t,\bar{\pi}_t) + \lambda[b - J_u^{(o)}(\pi_t,\bar{\pi}_t)]_+ - \min_{\bar{\pi}} \left(J(f,\pi_t,\bar{\pi}) + \lambda[b - J_u(\pi_t,\bar{\pi})]_+ \right) \\
&\le J_t^{(o)}(\pi_t,\bar{\pi}_t) + \lambda[b - J_u^{(o)}(\pi_t,\bar{\pi})]_+ - \min_{\bar{\pi}} \left(J^{(p)}(f,\pi_t,\bar{\pi}) + \lambda[b - J_u^{(p)}(\pi_t,\bar{\pi}_t)]_+ \right) \\
&\le J_t^{(o)}(\pi_t,\bar{\pi}_t) + \lambda[b - J_u^{(o)}(\pi_t,\bar{\pi}_t)]_+ - J^{(p)}(\pi_t,\bar{\pi}_t) - \lambda[b - J_u^{(p)}(\pi_t,\bar{\pi})]_+ \\
&\le |J_t^{(o)}(\pi_t,\bar{\pi}_t) - J(f,\pi_t,\bar{\pi}_t)| + |J^{(p)}(\pi_t,\bar{\pi}) - J(f,\pi_t,\bar{\pi}_t)| + \lambda[J_u^{(p)}(\pi_t,\bar{\pi}_t) - J_u^{(o)}(\pi_t,\bar{\pi}_t)]_+ \\
&\le |J_t^{(o)}(\pi_t,\bar{\pi}_t) - J(f,\pi_t,\bar{\pi}_t)| + |J^{(p)}(\pi_t,\bar{\pi}) - J(f,\pi_t,\bar{\pi}_t)| + \lambda|J_u^{(p)}(\pi_t,\bar{\pi}_t) - J_u^{(o)}(\pi_t,\bar{\pi}_t)| \\
&\le |J_t^{(o)}(\pi_t,\bar{\pi}_t) - J(f,\pi_t,\bar{\pi}_t)| + |J^{(p)}(\pi_t,\bar{\pi}) - J(f,\pi_t,\bar{\pi}_t)| + \lambda|J_u^{(p)}(\pi_t,\bar{\pi}_t) - J_u(f,\pi_t,\bar{\pi}_t)| + |J_u^{(o)}(\pi_t,\bar{\pi}_t) - J_u(f, \\
&\le L_{r,\pi} \sum_{h=0}^{H} \mathbb{E}\left[\|\mathbf{s}_{h,t} - \mathbf{s}_{h,t}^{(o)}\|_2 + \|\mathbf{s}_{h,t} - \mathbf{s}_{h,t}^{(p)}\|_2 \right] + \lambda L_{u,\pi} \sum_{h=0}^{H} \mathbb{E}\left[\|\mathbf{s}_{h,t} - \mathbf{s}_{h,t}^{(o)}\|_2 + \|\mathbf{s}_{h,t} - \mathbf{s}_{h,t}^{(p)}\|_2 \right]
\end{aligned}
\tag{24}
$$

Here we use Lemma 5, it follows that all the terms inside expectation are bounded in the same way as follows:

$$
\|\mathbf{s}_{h,t} - \mathbf{s}_{h,t}^{(o)}\|_2 \le 2\beta_{t-1} \left((1 + L_f + 2\beta_{t-1} L_\sigma) \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \right)^{h-1} \sum_{h'=0}^{h-1} \left\| \sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big) \right\|_2 ,
\tag{25}
$$

as $f^{(o)}$ and $f^{(p)}$ belong to the same plausible models $\mathcal{M}_t$. By applying the above inequality twice and denoting $C := \left((1 + L_f + 2L_\sigma) \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \right)$, we arrive at

$$
\begin{aligned}
\min_{\bar{\pi}} J(f,\pi^*,\bar{\pi}) - \min_{\bar{\pi}} J(f,\pi_t,\bar{\pi}) &\le 4(L_r + \lambda L_u)\beta_T^H C^H \sum_{h=0}^{H} \mathbb{E}\left[\sum_{h'=0}^{h-1} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2 \right], \\
&= 4L_{(r,\lambda,u)}\beta_T^H C^H \sum_{h=0}^{H} \mathbb{E}\left[\sum_{h'=0}^{h-1} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2 \right] \\
&\le 4L_{(r,\lambda,u)}\beta_T^H C^H H \sum_{h'=0}^{h-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2 \right]
\end{aligned}
\tag{26}
$$

where $t \le T$ and $1 \le \beta_t$ is non-decreasing in t.

□

Theorem 1. *Under Assumptions 1 to 4, let* $C = \left((1 + L_f + 2L_\sigma) \sqrt{1 + L_\pi^2 + L_{\bar{\pi}}^2} \right)$, $\Gamma_T = \sum_{t=1}^{T} \mathbb{E}\left(\left[\sum_{h'=0}^{H} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2^2 \right] \right)$ *and* $s_{h,t} \in \mathcal{S}$, $a_{h,t} \in$

$\mathcal{A}$ *and* $\bar{a}_{h,t} \in \bar{\mathcal{A}}$ *for all* $t, h > 0$*. Then for a fixed* $H \geq 1$*, with probability at least* $1 - \delta_r$*, the robust cumulative regret of RHC-UCRL is upper bounded by:*

$$R_T = O\left(L_{(r,\lambda,u)} \beta_T^H C^H H^{1.5} \sqrt{T\Gamma_T}\right) = \mathcal{O}\left(\sqrt{T}\right) \tag{27}$$

and with probability at least $1 - \delta_u$*, the robust cumulative violation of RHC-UCRL is upper bounded by*

$$V_T = O\left(L_u \beta_T^H C^H (1+\alpha) H^{1.5} \sqrt{T\Gamma_T}\right) = \mathcal{O}\left(\sqrt{T}\right) \tag{28}$$

Proof. Let us begin with the cumulative regret first

$$\begin{aligned}
R_T &= \sum_{t=1}^{T} \underbrace{\min_{\bar{\pi}\in\bar{\Pi}} J(f, \pi^*, \bar{\pi}) - \min_{\bar{\pi}\in\bar{\Pi}} J(f, \pi_t, \bar{\pi})}_{:=r_t} \\
&\leq \sqrt{T \sum_{t=0}^{T} r_t^2} \\
&\leq 4L_{(r,\lambda,u)} \beta_T^H C^H H \sqrt{T} \sqrt{\sum_{t=1}^{T} \left(\mathbb{E}\left[\sum_{h'=0}^{H-1} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2 \right] \right)^2} \\
&\leq 4L_{(r,\lambda,u)} \beta_T^H C^H H \sqrt{T} \sqrt{\sum_{t=1}^{T} \mathbb{E}\left(\left[\sum_{h'=0}^{H} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2 \right]^2 \right)} \\
&\leq 4L_{(r,\lambda,u)} \beta_T^H C^H H^{1.5} \sqrt{T} \sqrt{\underbrace{\sum_{t=1}^{T} \mathbb{E}\left(\left[\sum_{h'=0}^{H} \|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2^2 \right] \right)}_{\Gamma_T}}
\end{aligned} \tag{29}$$

The last inequality is due to the Cauchy-Schwarz inequality. And hence the theorem follows

Similarly, we now derive the same for the cumulative Violation bound

$$
\begin{aligned}
V_T &= \sum_{t=1}^{T} \underbrace{(b - \min_{\bar{\pi}\in\bar{\Pi}} J(f, \pi_t, \bar{\pi}))_+}_{v_t} \\
&\leq \sqrt{T \sum_{t=1}^{T} v_t^2} \\
&\sum_{h'=0}^{H-1} \mathbb{E}\left[\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right] \\
&\leq 2L_u H \beta_T^H C^H (1+\alpha)\sqrt{T} \sqrt{\sum_{t=1}^{T}\left(\sum_{h=0}^{H}\mathbb{E}\left[\sum_{h'=0}^{H}\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]\right)^2} \\
&\leq 2L_u H \beta_T^H C^H (1+\alpha)\sqrt{T} \sqrt{\sum_{t=1}^{T}\mathbb{E}\left(\left[\sum_{h'=0}^{H}\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2\right]^2\right)} \\
&\leq 2L_u \beta_T^H C^H (1+\alpha) H^{1.5}\sqrt{T} \sqrt{\underbrace{\sum_{t=1}^{T}\mathbb{E}\left(\left[\sum_{h'=0}^{H}\|\sigma_{t-1}\big(s_{h',t}, \pi_t(s_{h',t}), \bar{\pi}_t(s_{h',t})\big)\|_2^2\right]\right)}_{\Gamma_T}}
\end{aligned}
\tag{30}
$$

This completes the required proof. □

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